

Chapter: Introduction

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Exercise 1.9

Pattern Recognition and Machine Learning - Christopher M. Bishop

Solution

Part (a)

Let's show that the mode (i.e. the maximum) of the Gaussian distribution given by (1.46) is μ . Let's restate the equation (1.46) for clarity:

$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp \left\{ -\frac{1}{2\sigma^2}(x - \mu)^2 \right\} \quad (1.46)$$

To find the mode of the Gaussian distribution, we need to find the value of x that maximizes $\mathcal{N}(x|\mu, \sigma^2)$. Since the exponential function is monotonically increasing, we can focus on maximizing the exponent:

$$-\frac{1}{2\sigma^2}(x - \mu)^2 \quad (1)$$

To maximize this expression, we need to minimize $(x - \mu)^2$ because $-\frac{1}{2\sigma^2} < 0$. The minimum value of $(x - \mu)^2$ is 0 because $(x - \mu)^2 \geq 0$ for all x , which occurs when $x = \mu$. Therefore, the mode of the Gaussian distribution is μ and it is unique because $\sigma^2 > 0$.

Part (b)

Similarly, we can show that the mode of the multivariate Gaussian distribution given by (1.52) is $\boldsymbol{\mu}$. Let's restate the equation (1.52) for clarity:

$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \Sigma) = \frac{1}{(2\pi)^{D/2}|\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}) \right\} \quad (1.52)$$

where $\mathbf{x}$ is a D -dimensional vector, $\boldsymbol{\mu}$ is the mean vector, and Σ is the covariance $D \times D$ matrix and $|\Sigma|$ is the determinant of Σ .

To find the mode of the multivariate Gaussian distribution, we need to find the value of $\mathbf{x}$ that maximizes $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \Sigma)$. Similar to the univariate case, we can focus on maximizing the exponent:

$$-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}) \quad (2)$$

To maximize this expression, we need to minimize $(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})$. Since Σ is a positive definite matrix, Σ^{-1} is also positive definite, which means that $(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}) \geq 0$ for all $\mathbf{x}$. The minimum value of this expression is 0, which occurs when $\mathbf{x} = \boldsymbol{\mu}$. Therefore, the mode of the multivariate Gaussian distribution is $\boldsymbol{\mu}$ and it is unique because Σ is positive definite.

Justification. Since Σ is symmetric positive definite there exists an invertible symmetric square root $\Sigma^{1/2}$ with $\Sigma = \Sigma^{1/2}\Sigma^{1/2}$ and $\Sigma^{-1} = \Sigma^{-1/2}\Sigma^{-1/2}$. Hence

$$(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) = \|\Sigma^{-1/2}(\mathbf{x} - \boldsymbol{\mu})\|^2 \geq 0.$$

Equality holds iff $\Sigma^{-1/2}(\mathbf{x} - \boldsymbol{\mu}) = \mathbf{0}$, and since $\Sigma^{-1/2}$ is invertible this is equivalent to $\mathbf{x} = \boldsymbol{\mu}$.

Resources to learn more

- Square root of a positive definite matrix
- Cholesky decomposition